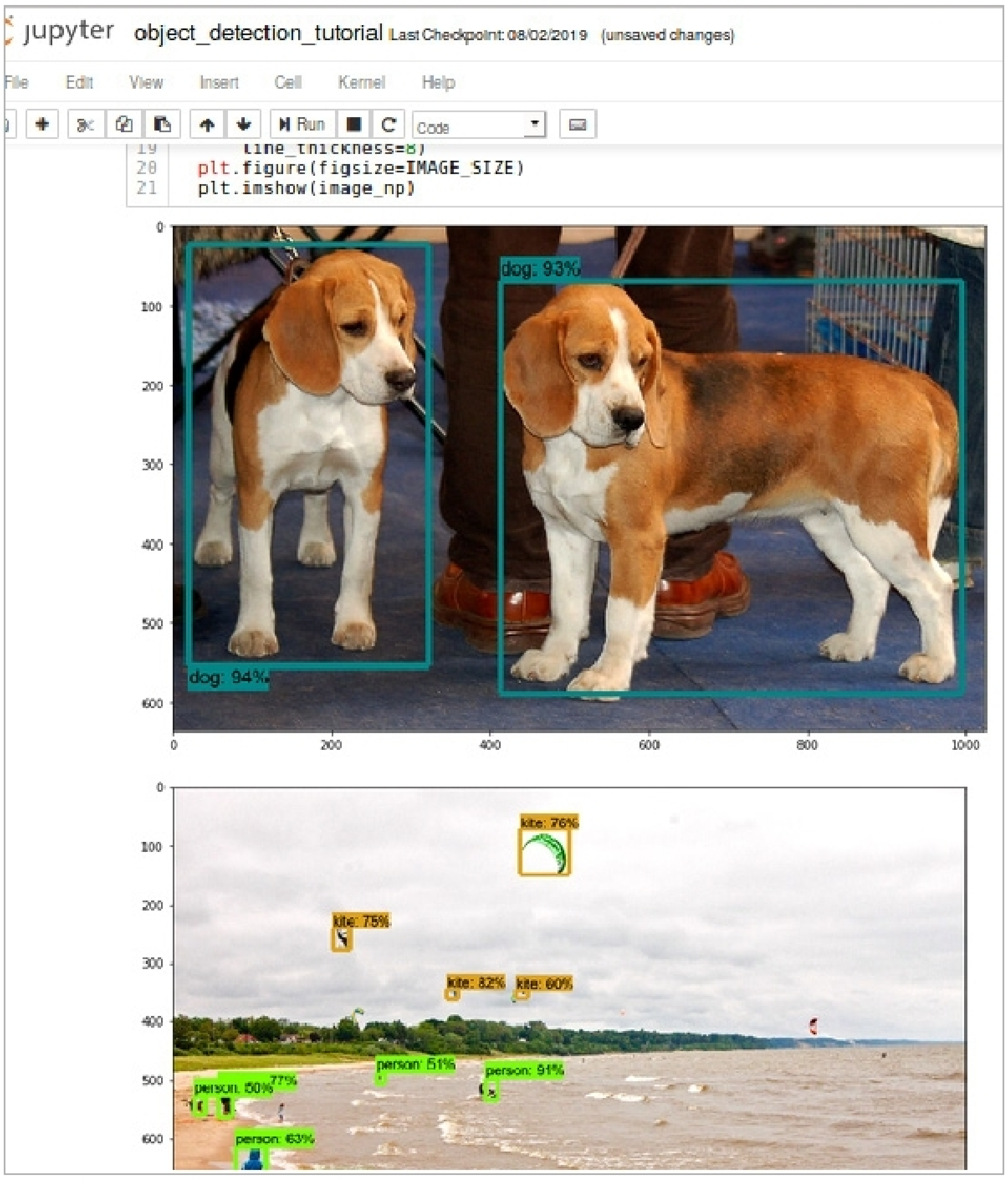

Figure 4: Object detection from an image

```
protoc object_detection/protos/anchor_generator.proto --python_out=.
```

Add libraries to *PYTHONPATH*. Run the following command from *tensorflow/models/research/*:

```
export PYTHONPATH=$PYTHONPATH:`pwd`:`pwd`/slim
```

You can test the installation of the TensorFlow Object Detection API by running the following command:

```
python object_detection/builders/model_builder_test.py
```

Run the object detection tutorial using the following command from the *tensorflow/models/research/object_detection* directory:

```
jupyter-notebook object_detection_tutorial.ipynb
```

It will run in the browser and will look like what's shown in Figure 3.

From the cell, select *Run all*, which will run and generate the output at the end of the page.

You will see the detection of all the images from the *test_images* directory (*tensorflow/models/research/object_detection/test_images*). If you want to test the code with your images, just add the path to the images, to the line *TEST_IMAGE_PATHS*.

```
for image_path in TEST_IMAGE_PATHS:
  image = Image.open(image_path)
```

If you want to add more images, you can change the number of images in the range, as follows:

```
PATH_TO_TEST_IMAGES_DIR = 'test_images'
TEST_IMAGE_PATHS = [ os.path.join(PATH_TO_TEST_IMAGES_DIR, 'image{}.jpg'.format(i)) for i in range(1, 5) ]

# Size, in inches, of the output images.
IMAGE_SIZE = (12, 8)
```

The final object identification will be shown as an output of images. The objects will be detected with labels, and the percentage/score of that object will be similar to the training set.

This very basic example can be followed for simple object identification. Moving objects/videos can also be tagged and identified in a similar manner, with a slight modification of code and addition of libraries like OpenCV, etc. END

References

[1] *https://en.wikipedia.org/wiki/Video_content_analysis*
[2] *https://www.tensorflow.org/*
[3] *https://www.nature.com/articles/nature14539*
[4] *https://www.edureka.co/blog/tensorflow-object-detection-tutorial/*

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